

# DESIGN AND TECHNICAL COMMISSIONING OF THE IN SITU NANOPROBE ENDSTATION AND INSTRUMENT AT THE ADVANCED PHOTON SOURCE\*

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## Abstract

The In Situ Nanoprobe (ISN) is a newly constructed, best-in-class experimental x-ray instrument at sector 19 of the upgraded Advanced Photon Source (APS-U). The new ISN beamline provides a 4.8-30 keV monochromatic x-ray beam, high coherence, and very high focused flux. Kirkpatrick-Baez (KB) mirror focusing offers a focal spot as low as 20 nm at a photon energy of 25 keV. The KB mirrors also provide a long working distance of 61 mm, to enable a versatile suite of sample environments: in-vacuum or in-air operation, heating to >1000 °C, cooling to 40 K, flow of liquids & gases, and applied electrical fields. The instrument supports fast fly-scanning of relatively large and heavy samples of ~10x10 mm and 2 kg at 1 mm/s. Measurement techniques include 2D and 3D x-ray fluorescent mapping, ptychographic coherent structural imaging, x-ray diffraction, x-ray excited optical luminescence and x-ray beam induced current and voltage measurements. This work presents the first mechanical results from the ongoing technical commissioning in Summer 2025, including the design and architecture of the endstation, vibrational and thermal management, beam conditioning, KB mirror design, vacuum chamber design, sample scanning, metrology, sample environments and detector systems design.

## INTRODUCTION

The Advanced Photon Source (APS) at Argonne National Laboratory (ANL) has recently completed a major upgrade project (APS-U), increasing the brightness of the x-ray beams by up to 500 times with world-record beam emittance / coherence [1]. This upgrade will enable scientific discoveries across many disciplines.

The In Situ Nanoprobe (ISN) is a hard x-ray beamline [2] that was newly constructed, among many other APS-U beamlines, to take full advantage of the increased brightness and coherence. ISN is designed to operate with a very stable monochromatic x-ray beam of 4.8 to 30 keV. ISN was envisioned to study complex materials systems under *in-situ* and *operando* conditions across many lengths scales. This work details the numerous advancements in endstation and instrument engineering necessary to realize this goal and reports the first mechanical results from the ongoing technical commissioning.

\* This research used resources of the Advanced Photon Source, a U.S. Department of Energy (DOE) Office of Science User Facility operated for the DOE Office of Science by Argonne National Laboratory under Contract No. DE-AC02-06CH11357

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## ENDSTATION ENCLOSURE

The new ISN long beamline (see Fig. 1) is located at Sector 19-ID of APS and the ISN instrument is located in a satellite building at a distance of 220 m from the undulator source. The endstation enclosure is designed to isolate ambient and transient vibrations and mitigate thermal drifts, with the goal of maintaining stability of the nanofocusing mirrors and ultimately the x-ray focal spot.



Figure 1. ISN beamline and satellite building at APS, outlined in black.

Thermal stability is achieved with a passive system comprised of an intermediate air environment, vestibule, and laminar flow air socks (see Fig. 2). All HVAC supply and exhaust piping are routed through labyrinths in the ceiling and a vibration isolated exhaust fan is outside the enclosure. Hutch air temperature must stay within  $\pm 0.05$  °C for 1 hr and the instrument surface temperature within  $\pm 0.01$  °C for 1 hr.

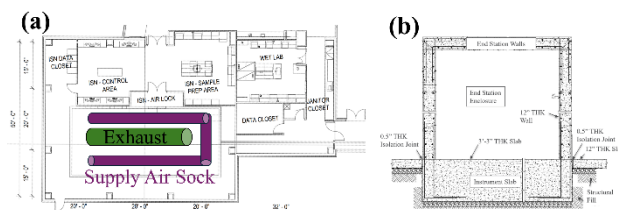


Figure 2. (a) Schematic layout of the ISN satellite building and endstation. (b) Cross-section view of the end station concrete foundation, isolation joint, and walls.

Vibrational stability is aided by a stand-alone concrete slab of thickness 1 m, with isolation joints damping from the surrounding floor. These construction specifications were made to meet a vibration criteria of VC-G, or 0.78  $\mu\text{m/s rms}$  across 1-100 Hz band, based on similar facilities [3].

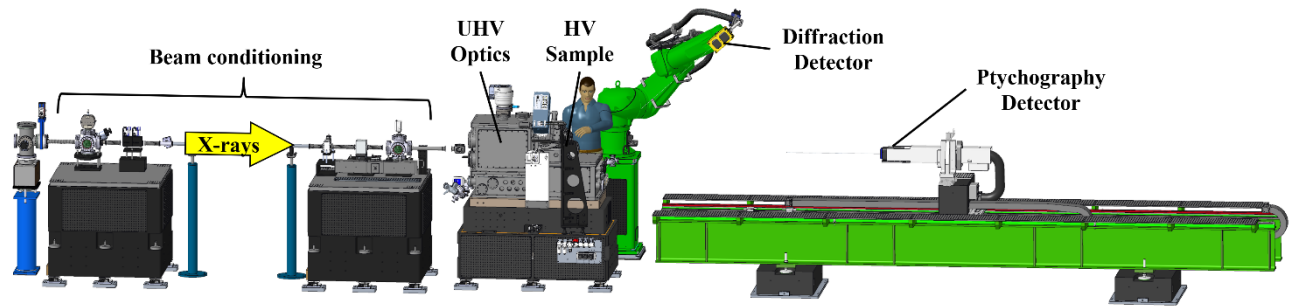


Figure 3. Layout of all major components in the ISN endstation.

## BEAM CONDITIONING SECTION

The endstation (see Fig. 3) houses beam conditioning components upstream of the instrument, including two scintillator beam position monitors (BPMs), x-ray filters for attenuation, clean up slits, a fast shutter, two pin diodes for flux measurement, and a square pinhole after the KB mirrors. All components are mounted on large granite blocks for stability and most components operate in low vacuum.

## INSTRUMENT DESIGN

The ISN instrument core (see Fig. 4) is comprised of both nano-focusing optics and sample scanning stages that share a common invar baseplate but are housed in separate vacuum chambers. This enables the KB optics to remain in a clean UHV state while allowing frequent venting and loading of samples and retains a relatively athermal connection for metrology. Both vacuum chambers have single crystal diamond windows on both upstream and downstream sides, with a 2 mm helium gap between the chambers.

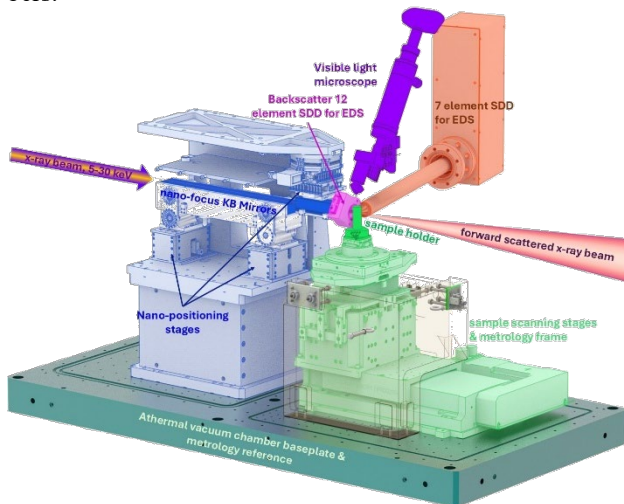


Figure 4. ISN instrument with vacuum chamber hidden.

The instrument is supported on a highly stable granite table, with a motorized air bearing surface for coarse in-board/outboard and yaw alignment. When de-pressurized, the table behaves as monolithic rigid structure [4].

Kirkpatrick-Baez (KB) mirrors are developed by JTEC are implemented, with a designed focal spot of 20 nm at 25 keV and a large working distance of 61 mm. The mirror

positions and orientations are controlled with six custom piezo-driven flexure-guided nanopositioning stages [5], mounted on a rigid invar structure. We require nrad level angular stability to maintain their relative position within 20% of the focus size (see Fig. 5). The mirrors were aligned after installation with a combination of a CMM arm, auto-collimator, and x-ray techniques.

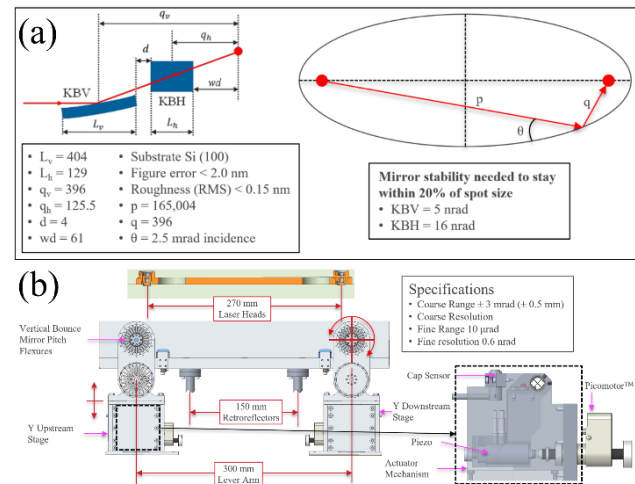


Figure 5. (a) KB mirror specifications (b) Vertical bounce KB mirror, stages, and interferometer sensor positions.

The sample is fly scanned using an XYZ stage stack with 10 mm<sup>3</sup> travel range. An Aerotech PRO225SL stage is used along the beam direction and Aerotech ANT130L along the transverse horizontal direction. An APS-designed vertical flexure stage with combination of stepper motor & piezo actuators is used for coarse/fine travel. The internal piezo actuators double as tip/tilt correction. Fly-scan speeds depend on the size of the region-of-interest (ROI) and the direction of fly-axis. For vertical fly scan with ROI size <90 μm, the maximum speed is 1 mm/s. For vertical fly-scan with ROI size <10 mm, the maximum speed is 0.1 mm/s. All horizontal fly-scans have a maximum speed of 1 mm/s.

A Micronix PR-160 rotation stage is used for 3D tomography, ptychography and diffraction. The runout and wobble errors of the rotation stage are measured with capacitive sensors probing an aluminium reference ring with diamond-turned cylinder and flat surfaces with 20 nm precision. The wobble and runout are then corrected by the XYZ stages.

The KB optics positions are measured with respect to the sample using a 15-channel interferometer system and internal/external Zerodur and Invar reference frames.

## SAMPLE ENVIRONMENTS

The instrument sample chamber can operate in high vacuum, air, or inert gas backfill modes, with large space claim volumes reserved all around, apart from detector snouts. ISN sample holders mount on top of the scanning stage stack with a quasi-Maxwell kinematic coupling and magnetic pre-load and mounting surface of 60 mm below the x-ray beam elevation. The maximum size envelope is a 24 mm diameter cylinder, assuming all detectors will be used, and the sample will be rotated. Fast scanning performance is expected to sustain with a payload up to 2 kg. Several large ports are available on the instrument chamber for multiple purposes: reaching gloved hands inside to mount a holder, observing, and connecting feedthroughs for fluid, electrical, etc.

In addition to simple pin and flag mount sample holders, the beamline offers three advanced in situ sample holder systems, along with a full suite of sample preparation and operating equipment. All three systems allow the sample to rotate over large angles for tomography.

The first advanced in situ sample holder (developed by Hummingbird Scientific) is for electrochemical experiments, with two inlet fluid lines, heating beyond 100 °C, bulk electrodes, and >200  $\mu\text{m}$  thick flow cell for sample particles.

The second holder (developed by Hummingbird Scientific) is for catalysis and materials synthesis experiments with two inlets for pressure-controlled gases, four channels of electrical biasing, and heating up to 1000 °C with low thermal drift and excellent thermal uniformity.

The third sample holder (still in development by APS) is a variable temperature system for quantum and condensed matter research, with feedback-controlled cryogenic sample cooling from 40 K and heating up to 355 K, via an ultra-low vibration, closed-cycle helium cryostat (ARS Cryo).

## DETECTORS

ISN is equipped with a suite of detector systems for simultaneously multimodal imaging. 2D and 3D x-ray fluorescence (XRF) measurements are conducted with either a Hitachi 7 element silicon drift detector (SDD) or a Ray-Spec 12 element backscatter SDD, with Xpress3 MK2 readout electronics.

Wide-angle x-ray diffraction (XRD) measurements are conducted with an Eiger2 X 1M photon counting detector, carried by a Stäubli TX2-160L industrial robot to position in spherical coordinates. The signal is transmitted through a large domed beryllium window (developed by Materion). The beryllium window is 0.5 mm thick, is mounted at a complex angle of 10° above the x-ray beam elevation plane and 25° outboard, has a clear aperture of 152 mm, and transmits a solid angle of  $\sim 0.25$  sr.

A second Eiger2 X 1M is mounted on a downstream XYZ motion table with 6 m travel range along the beam direction for forward scattering ptychography, small angle

scattering, or absorption measurements. X-ray beam induced voltage/current measurements are conducted with a Caen Els TetrAMM electrometer.

A custom in-line vacuum microscope (IVM) is installed just upstream of the sample, with a NIR 10X/0.26NA infinity corrected objective and mounted on XYZ alignment stages. A fold mirror with a hole in the optical path allows the x-ray beam to pass through to the sample, enabling x-ray imaging in parallel with optical imaging. The IVM system has a field view of 600  $\mu\text{m}$  and a resolution of 1  $\mu\text{m}$ . A motorized optical path selector enables partial or full collection of 430 - 1800 nm emission from the sample. A reflective collimator intercepts the collected emission and passes it via optical fiber to a spectrometer. An external optical light injection module (Optique Peter) sends excitation or stimulation light to the sample. This IVM enables x-ray excited optical luminescence microscopy.

The full suite of data from ISN can be rapidly transferred and processed via a data pipeline to the Aurora exascale supercomputer at the Argonne Leadership Computing Facility [6].

## INITIAL COMMISSIONING RESULTS

Ongoing technical commissioning shows promising results. Preliminary KB mirror stability was measured at 0.4 nrad RMS and 1.7 nrad peak-to-peak noise (See Fig. 6), achieving the <5 nrad stability requirement. Preliminary focal spot size measurements also show promise of achieving the ultimate design goal of 20 nm.

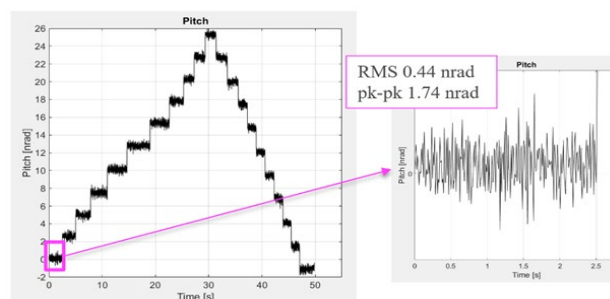


Figure 6. Pitch motion of the vertical KB mirror with  $\sim 2$  nrad steps. Measurement conducted with fiber interferometer heads operating in-air and vacuum pumps off.

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